

ISHU KUMAR

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Senior Backend & AI Systems Engineer | Python | Django | Distributed Systems | Agentic AI

SUMMARY

Senior Backend & Distributed Systems Engineer with 7+ years of experience designing high-scale APIs, multi-tenant cloud platforms, and production AI systems. Specialized in Python (FastAPI/Django), Azure/AWS distributed architectures, event-driven pipelines, and LLM orchestration (MCP, RAG, tool calling), with hands-on multi-agent system development.

TECHNICAL SKILLS

Primary: Python, Django, FastAPI, Django REST Framework, PostgreSQL, Azure, AWS, Distributed Systems, REST APIs, Event-Driven Architecture

Backend & Architecture: Flask, WebSockets, asynchronous processing, multi-tenancy, microservices, modular monoliths, serverless, message queues, pub/sub, caching, concurrency, service-to-service communication

Cloud & Infrastructure: OpenStack, VMware, GCP, Docker, Terraform, CloudFormation, CI/CD, Linux, Git

Data: MySQL, Redis, Oracle Database, Amazon Redshift, Supabase, LanceDB

AI & Agentic Systems: RAG, MCP, AI agents, multi-agent orchestration, tool calling, agent workflows, vector search, LLM integrations, OpenAI, Anthropic, Ollama, Oracle AI Vector Search

Testing & Operations: Pytest, unittest, Playwright, Vitest, integration/API/E2E testing, load/performance testing, Grafana, Loki

Languages: English (Professional / Fluent), Hindi (Native)

EXPERIENCE

ImpplyCloud

Oct 2025 – Mar 2026

Technical Lead

- Led backend modernization of the AnnexCloud multi-tenant e-commerce SaaS platform serving **15M+ end users across 4 enterprise tenants**, migrating core functionality from PHP to Python on Azure with PostgreSQL/MySQL, Redis, event-driven processing, and serverless components.
- Re-engineered the legacy PHP user-management core into modular Python services and integrated transactions, rewards, points, tiers, milestones, and related product domains, supporting the platform's 15M+ user base.
- Designed event-driven data propagation using REST APIs, Azure Service Bus, Azure Functions, PostgreSQL/MySQL, and external webhooks; processed and routed persisted data to the correct endpoint with acknowledgement-based delivery confirmation.
- Established Azure Functions, messaging infrastructure, Kafka-based workflows, caching, Celery processing, and CI/CD deployment; implemented authentication and production deployment workflows.
- Redesigned user-management and rewards workflows with asynchronous execution and established Grafana/Loki observability, reducing downtime incidents by **40%** and incident response time by **25%**.
- Mentored 6 engineers across Python architecture, system design, secure coding, and delivery.

Oracle India

May 2025 – Oct 2025

Senior Software Developer

- Implemented **Model Context Protocol (MCP)** capabilities in Oracle AI Optimizer, including an MCP server for vector-search functionality and MCP client capabilities enabling the application to operate as an MCP client.
- Engineered Python APIs and low-latency retrieval integrations with Oracle Database and Oracle AI Vector Search, supporting database-backed RAG workflows and asynchronous processing.
- Implemented RAG, Vector Store, tool-integration, and testing functionality across AI application workflows, establishing standardized MCP-based tool interfaces for LLM applications.

CloudBolt Software

Dec 2021 – Mar 2025

Senior Software Engineer

- Engineered backend capabilities for a large Django/DRF cloud-management platform spanning AWS, Azure, GCP, OpenStack, VMware, and other infrastructure providers.
- Built resource handlers, environments, instances, volumes, SSH keys, authentication/authorization panels, Django admin interfaces, and APIs for cloud-resource creation, deployment, lifecycle management, and synchronization.
- Engineered synchronization workflows supporting **1,000+ child accounts under a single parent account** for AWS and Azure, synchronizing accounts, instances, environments, histories, and cloud resources.
- Architected multi-threaded processing and Redis caching across Django ORM/PostgreSQL workflows, improving resource retrieval, synchronization, and API performance across enterprise cloud-management operations.
- Designed orchestration workflows for EKS, ECS, and corresponding services across cloud providers, including real-time deployment updates and asynchronous resource synchronization.
- Worked directly with external enterprise customers to reproduce production issues, translate requirements into technical designs, support deployments, and participate in technical discussions and demos.
- Implemented stale-resource detection and cloud-cost optimization workflows, contributing approximately **\$3,000/month in customer savings**; maintained **80%+ test coverage**.
- Resolved complex multi-cloud SDK upgrades across AWS, Azure, and OpenStack by reconciling breaking API/library changes, deprecated interfaces, security changes, and version inconsistencies; built a custom Azure billing module where the available

SDK/billing implementation did not provide the required information.

Deloitte

Apr 2021 – Nov 2021

Analyst

- Automated a multi-workbook Excel data-processing pipeline using Python and VB, consolidating data from multiple sheets into standardized output workbooks and automating the workflow end-to-end.
- Reduced processing time for recurring workbook workflows from hours of manual work to approximately **10–30 minutes per workbook**, representing roughly **90% reduction** for the targeted process.
- Reduced large-scale analytics runtime by **40%** using PySpark and Amazon Redshift.

Root Info Solutions

Sep 2020 – Apr 2021

Associate Software Engineer

- Built backend services for a real-time mobile platform supporting **50,000+ users**, using Django, DRF, PostgreSQL, Django Channels, Redis, Celery, RabbitMQ, ASGI, and Nginx.
- Implemented WebSocket-based real-time communication for chat, groups, typing notifications, driver location updates, bidding, delivery, wallet, and payment workflows.
- Developed APIs and backend services across multiple domains and worked directly with external enterprise customers on requirements, troubleshooting, implementation, deployment, and production support.

Skysoft Global

Oct 2019 – Sep 2020

Python / Django Software Engineer

- Built multi-tenant hospital inventory and scheduling systems using Python, Django, DRF, PostgreSQL, and Django tenant schemas for hospital chains and multiple owners.
- Designed shared and tenant-specific data models for inventory, doctor rosters, leaves, appointments, and patient booking; enabled doctors to declare leave before roster generation so unavailable slots were automatically excluded.
- Built a gamified institutional collaboration platform with wallet/rewards functionality for discussions, events, and information exchange; also developed Docubank, a document-management application with OCR, organized document collections, Razorpay payments, Django/DRF, and PostgreSQL.

PROJECTS & OPEN SOURCE

AI Education / MCQ Generation & Validation – 5-week Engineering Engagement

Designed and implemented a multi-agent system for real-time MCQ generation and validation across student levels, orchestrating specialized **Generator, Evaluator, Validator, Verifier, and Finalizer agents**. Built validation workflows for answer correctness, ambiguity and multiple-best-answer detection, question similarity, content quality, and student-level alignment. Designed and implemented the agent architecture and coordination hands-on while using AI coding tools to accelerate development.

VaultMind – Local-First RAG Knowledge System

Built a local-first RAG knowledge system with LanceDB and Ollama, including conversational workflows, tool calling, agent-oriented task execution, LLM integrations, and database-backed retrieval.

Zocept – AI Credit-Card Rewards Product

Own Product

- Built and operated the product end-to-end using Vite/Tailwind, Supabase/PostgreSQL, Redis, OpenAI, Razorpay, OAuth, OTP authentication, Resend, Vercel, and GitHub Actions.
- Covered 213 credit cards across major Indian banks, implemented recurring Razorpay subscriptions, and created **516 automated tests**, including 358 Playwright E2E and 158 Vitest unit tests.

eStore Manager – AI Sales Automation

Production AI System

- Designed an end-to-end workflow for **100,000+ users** that receives order/enquiry emails, builds and queries a RAG database from store inventory, places orders, and generates AI responses within approximately **15–30 seconds**.

AI VM Analyzer – Cloud Optimization

Production AI System

- Built a workflow that collects cloud metrics, analyzes VM utilization, generates reports and AI summaries, and recommends stale/deletable or under/over-provisioned resources, delivering **30%+ cost savings per customer**.

Oracle AI Optimizer / MCP: MCP server/client implementation and Oracle AI Vector Search integration contributed as part of Oracle's AI Optimizer open-source work.

MCP ChatBot: Multi-provider MCP client connecting one chat workflow to multiple LLM providers.

EDUCATION

M.Tech, Computer Science and Engineering – Central University of Rajasthan

2017–2019

B.Tech, Civil Engineering – Rajkiya Engineering College, Bijnor

2011–2015

CERTIFICATIONS

AI Agents & Agentic AI with Python | MCP: Rich-Context AI Apps (Anthropic) | AWS Serverless
CloudBolt Certified CMP Engineer | OWASP Top 10:2021 in Python